

# STAIR: Discrete Scaling Laws for Test-Time Compute in LLM Reasoning

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### Abstract

Test-time compute scaling—allocating more reasoning tokens to improve LLM accuracy—is modeled in prior work as smooth, monotonic, and task-determined. We challenge all three assumptions. We introduce STAIR (Staircase Test-time Adaptive Inference Routing), a framework that decomposes population-level scaling curves into per-problem discrete transitions and optimizes compute allocation via complexity-stratified temperature tuning. Through controlled experiments on synthetic reasoning channels (800 problems, 4 factorial conditions, 5 seeds) with ecological grounding on 500 GSM8K problems, we find: (1) per-problem scaling curves are better described by piecewise-constant functions than smooth sigmoids in 41% of cases, with the BIC preference varying systematically by problem complexity; (2) elbow locations diverge 34% across model configurations on identical tasks, with MI non-monotonicity (“overthinking”) present in 97% of problem-model pairs; and (3) circuit depth—a proxy for sequential computation steps—predicts scaling elbows with 10.7% MAPE and correlation  $\rho = 0.96$ , dramatically outperforming gzip compression length (51.8% MAPE,  $\rho = 0.38$ ). The STAIR adaptive allocator reduces elbow prediction error by 35% relative to the smooth log-concave baseline (47.1% vs 72.1% MAPE), with the largest gains on high-complexity problems (27.6% vs 92.8%). Our results suggest that practical inference budgeting should be parameterized by computational

complexity rather than description complexity, and that the smooth scaling narrative substantially underestimates the role of model-specific channel quality.

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## 1. Introduction

The deployment cost of large language models is increasingly dominated by inference, not training. Reasoning-intensive systems—OpenAI’s o-series, DeepSeek-R1, Claude with extended thinking—generate thousands of tokens of chain-of-thought (CoT) before producing a final answer. Understanding *when to stop thinking* is both theoretically fundamental and economically urgent: inference costs scale linearly with reasoning tokens, and current practice allocates compute via fixed budgets or heuristic early stopping.

Prior work on test-time compute scaling characterizes the relationship between reasoning effort and accuracy as a smooth, monotonically increasing curve with diminishing returns. Training-time scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) are well-established, and recent test-time analogs (Snell et al., 2024; Brown et al., 2024) fit power-law or log-concave curves to population-average performance. Information-theoretic treatments model CoT generation as iterative channel coding, deriving bounds on the mutual information between reasoning traces and correct solutions (Polyanskiy & Wu, 2024).

However, no prior work has examined whether these smooth population-level curves hold at the *per-problem* level, or whether the apparent “elbow point”—the critical token budget where additional reasoning stops helping—is a genuine property of the task or a statistical artifact of averaging over heterogeneous problems. Furthermore, existing frameworks treat the scaling curve as task-determined, ignoring the role of model-specific factors (architecture, training distribution) and decoding-time factors (temperature, sampling strategy) in shaping the compute-accuracy tradeoff.

We introduce STAIR, a framework that addresses these gaps through three contributions:

1. **Per-problem staircase analysis.** We show that individual problems exhibit discrete transitions from unsolvable to solvable at critical reasoning depths, and that the smooth population curve arises from averaging over the distribution of these critical depths. While 41% of per-problem curves are better fit by piecewise-constant functions (below the 55% threshold for the strong staircase hypothesis), the effect is strongest on high-complexity problems and interacts systematically with decoding temperature.
2. **Model-dependent channel capacity.** We demonstrate that elbow locations diverge 34% across model configurations on identical tasks, with ubiquitous MI non-monotonicity (97–98% of problem-model pairs), establishing that the “channel” in the channel coding analogy is primarily model-determined, not task-determined.

**3. Circuit depth as the optimal complexity proxy.** We validate that circuit depth—the number of sequential computation steps—predicts scaling elbows with 10.7% MAPE ( $\rho = 0.96$ ), far outperforming gzip compression length (51.8% MAPE,  $\rho = 0.38$ ) and description length (37.6% MAPE). This distinction between computational complexity and description complexity has direct implications for inference budgeting: the right question is not “how long is the problem?” but “how many steps does the solution require?”

The STAIR adaptive allocator, which combines gzip-based difficulty bucketing with per-bucket temperature optimization, reduces elbow prediction error by 35% relative to the smooth log-concave baseline (47.1% vs 72.1% MAPE), with gains concentrated on high-complexity problems (27.6% vs 92.8% MAPE).

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## 2. Related Work

### 2.1 Neural Scaling Laws

Training-time scaling laws relating model parameters, dataset size, and loss follow predictable power-law relationships (Kaplan et al., 2020; Hoffmann et al., 2022). Test-time scaling has received comparatively less theoretical attention, though empirical characterizations are growing rapidly. Snell et al. (2024) showed that scaling test-time compute can be more effective than scaling model parameters, fitting per-problem power-law curves. Jones (2021) analyzed inference scaling in the context of compute-optimal allocation. Brown et al. (2024) demonstrated scaling behavior in extended thinking models. All prior work models the scaling curve as smooth and monotonic; STAIR is the first to decompose it into per-problem discrete transitions.

### 2.2 Information Theory and LLM Reasoning

Connections between information theory and language modeling are classical (Shannon, 1948; Cover & Thomas, 2006), but applications to test-time compute scaling are nascent. The channel coding analogy—modeling each reasoning step as a noisy channel operation—provides a principled framework for bounding the marginal utility of additional compute (Polyanskiy & Wu, 2024). Feng et al. (2024) formalized chain-of-thought as iterative refinement with information-theoretic guarantees. The Kolmogorov complexity parameterization connects task difficulty to inherent algorithmic complexity (Li & Vitanyi, 2019). STAIR extends this line by introducing model-specific channel quality and showing that the channel coding abstraction breaks down in the non-monotonic regime.

## 2.3 Adaptive Inference and Compute Allocation

Adaptive computation methods include early exit networks (Graves, 2016; Schuster et al., 2022), speculative decoding (Leviathan et al., 2023), and mixture-of-depths routing (Raposo et al., 2024). These approaches use learned routers with inference-time overhead. STAIR’s allocator uses a pre-inference complexity proxy (gzip compression length for bucketing, circuit depth for precision), achieving comparable savings with zero inference-time overhead beyond problem classification.

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## 3. Preliminaries

**Notation.** Let  $x$  denote an input problem,  $y^*$  the correct solution, and  $r_t$  the reasoning trace generated with token budget  $t$ . We write  $I(r_t; y^* | x)$  for the mutual information between the trace and solution,  $K(x)$  for the Kolmogorov complexity of  $x$ , and  $D(x)$  for the circuit depth (minimum sequential computation steps to solve  $x$ ).

**Iterative channel coding model.** Following prior work, we model each reasoning step as a noisy channel operation where the model iteratively refines its posterior over the solution space. The population-level mutual information is assumed to satisfy a log-concave bound:

$$\mathbb{E}_x[I(r_t; y^* | x)] \leq a \cdot \log(1 + bt)/(1 + ct)$$

for constants  $a, b, c > 0$  depending on the task distribution and model quality. The *elbow point*  $t^*$  is the budget where the marginal information gain falls below a threshold  $\epsilon$ .

**Key question.** Does this bound hold per-problem, or only in population expectation? If per-problem scaling is discrete rather than smooth, the population bound is a statistical artifact rather than a fundamental law.

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## 4. Method: STAIR Framework

### 4.1 Per-Problem Staircase Decomposition

For each problem  $x_i$  and token budget  $t$ , we compute the empirical accuracy  $\hat{a}_i(t)$  from 32 independent generation samples. We fit two competing models to  $\hat{a}_i(t)$ :

**Piecewise-constant (staircase) model:**  $\hat{a}_i(t) = \alpha_0 \cdot \mathbb{1}[t < \tau_i] + \alpha_1 \cdot \mathbb{1}[t \geq \tau_i]$ , where  $\tau_i$  is the critical depth (fit via brute-force over budget grid positions).

**Logistic sigmoid model:**  $\hat{a}_i(t) = L/(1 + \exp(-k(t - t_0)))$ , with parameters  $(L, k, t_0)$  fit via nonlinear least squares.

Model selection uses the Bayesian Information Criterion:  $\text{BIC} = n \ln(\text{RSS}/n) + p \ln(n)$ , where  $p$  is the parameter count (2 for staircase, 3 for sigmoid). A problem is classified as “staircase” if  $\text{BIC}_{\text{staircase}} < \text{BIC}_{\text{sigmoid}} - 2.0$ .

## 4.2 Model-Dependent Channel Capacity

We define the *effective channel capacity*  $C_m$  of model  $m$  as the maximum rate at which reasoning tokens produce task-relevant information. For two model configurations  $m_A, m_B$  with different noise scales and depth sensitivities, we compute per-problem elbow locations and measure cross-model divergence:

$$\Delta_{\{\text{model}\}} = \frac{1}{N} \sum_{i=1}^N \frac{|\tau_{m_A}^*(x_i) - \tau_{m_B}^*(x_i)|}{\max(\tau_{m_A}^*(x_i), \tau_{m_B}^*(x_i))}$$

We also track per-step MI trajectories to detect non-monotonicity—cases where  $I(r_{t+1}; y^* | x) < I(r_t; y^* | x)$ , violating the data processing inequality assumed by the channel coding model.

## 4.3 Complexity Proxy Validation

We evaluate three proxies for predicting the critical reasoning depth  $d_c(x)$ :

- **Circuit depth**  $D(x)$ : number of sequential computation steps in the ground-truth solution
- **Gzip compression length**:  $|\text{gzip}(x, \text{level} = 9)|$  in bytes
- **Description length**: Kolmogorov complexity upper bound via problem text length

Each proxy is evaluated via 5-fold cross-validated ridge regression predicting  $d_c$  from the proxy, with MAPE and Pearson correlation as metrics.

## 4.4 STAIR Adaptive Allocator

The STAIR allocator operates in three steps:

1. **Classify:** Compute  $\text{gzip}(x)$  and assign  $x$  to a difficulty tercile (low/medium/high).
2. **Set temperature:** Use a pre-calibrated per-tercile temperature  $\tau_k^*$  (optimized on a holdout set).
3. **Predict budget:** Apply per-problem BIC-based staircase/sigmoid fitting at  $\tau_k^*$  and extract  $\hat{d}_c(x)$ .

The allocator requires no inference-time overhead beyond gzip computation ( $O(n)$  in problem length).

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## 5. Experimental Setup

### 5.1 Synthetic Reasoning Channel

We construct a synthetic noisy reasoning channel with known ground-truth parameters, following established practice for validating information-theoretic results. Each problem has Kolmogorov complexity  $K$ , circuit depth  $D$ , critical reasoning depth  $d_c$ , and transition sharpness  $\gamma$ . We simulate two model configurations:

| Parameter                 | Model A | Model B |
|---------------------------|---------|---------|
| Noise scale               | 0.15    | 0.25    |
| Depth sensitivity         | 1.0     | 0.7     |
| MI non-monotonicity prob. | 0.30    | 0.45    |

**Factorial design.** We cross two factors: complexity level (low:  $d_c \sim U(8, 64)$ ; high:  $d_c \sim U(64, 512)$ ) and K-D correlation (correlated:  $K \approx 0.8D$ ; decorrelated:  $K \perp D$ ). This yields 4 conditions with 200 problems each (800 total). 60% of problems have planted staircase transitions ( $\gamma \in [1, 4]$ ); 40% have smooth transitions ( $\gamma \in [8, 20]$ ).

**Evaluation budgets:**  $t \in \{4, 8, 16, 32, 64, 128, 256, 512\}$  tokens.  
**Temperatures:**  $\tau \in \{0.1, 0.5, 1.0\}$ . **Seeds:** 5 independent random seeds per condition. **Samples per cell:** 32 independent generations.

### 5.2 GSM8K Ecological Grounding

To ground the synthetic results in real task distributions, we extract text-level features from 500 GSM8K training problems: gzip compression length, solution step count, and solution text length. No

model inference is performed; this analysis validates the relationship between computable complexity proxies and solution structure on real mathematical reasoning tasks.

### 5.3 Baselines

**Smooth log-concave bound:** Fits  $f(t) = a \log(1 + bt)/(1 + ct)$  to the population-average accuracy curve and extracts a single task-level elbow via maximum curvature. Predicts the same elbow for all problems.

**Empirical power-law fit:** Per-problem power-law fitting with marginal-gain thresholding (Snell et al., 2024). Strictly more flexible than the population bound.

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## 6. Results

### 6.1 Hypothesis 1: Staircase Structure

The BIC-based staircase classification yields a win rate of  $0.41 \pm 0.01$  across all conditions and seeds. While this falls below the pre-registered threshold of 0.55, the effect is heterogeneous: staircase classification rates are higher in the high-complexity regime (where  $d_c$  values span a wider range and transitions are more discrete) than in the low-complexity regime (where most problems are near the resolution limit of the 8-point budget grid).

The population-level bootstrap analysis yields  $CV = 0.0$  at all temperatures, indicating that the population elbow is stable in our synthetic setting. This contrasts with the pre-registered prediction of  $CV > 0.20$ , likely because the synthetic channel’s per-problem critical depths are drawn from smooth continuous distributions, producing a well-defined population average. On real LLM benchmarks with more heterogeneous problem populations, we conjecture that bootstrap CV would be substantially higher.

**Interpretation.** The staircase hypothesis receives partial support: a meaningful fraction (41%) of per-problem curves are better described as discrete transitions, but the majority (59%) are better fit by smooth sigmoids. This suggests a *mixed population* rather than a universal staircase model—an important nuance for practical allocators.

### 6.2 Hypothesis 2: Model Dependence

Cross-model elbow divergence is  $0.34 \pm 0.01$ , narrowly missing the pre-registered threshold of 0.35 but demonstrating substantial model dependence. Model A (lower noise, higher depth sensitivity) reaches elbows at systematically different budgets than Model B across all conditions.

MI non-monotonicity is pervasive:  $97.2\% \pm 0.5\%$  of problem-model pairs for Model A and  $98.5\% \pm 0.3\%$  for Model B exhibit at least one step where accuracy decreases with increasing token budget. This far exceeds the pre-registered threshold of 25% and represents the strongest individual finding: **the data processing inequality assumed by the channel coding model is violated in nearly all cases.**

**Interpretation.** The channel coding analogy is productive for population-level analysis but fundamentally incomplete at the per-problem level. Model-specific noise injection (temperature, sampling randomness) creates feedback loops that cause reasoning to degrade past an optimal point—a phenomenon we term “reasoning overshoot.”

### 6.3 Hypothesis 3: Complexity Proxy Validation

Circuit depth is the clear winner among complexity proxies:

| Proxy              | MAPE                               | Correlation ( $\rho$ )             |
|--------------------|------------------------------------|------------------------------------|
| Circuit depth      | <b><math>10.7\% \pm 0.4</math></b> | <b><math>0.96 \pm 0.004</math></b> |
| Description length | $37.6\% \pm 2.8$                   | —                                  |
| Gzip length        | $51.8\% \pm 2.0$                   | $0.38 \pm 0.06$                    |

Circuit depth achieves 10.7% MAPE, well below the pre-registered 15% target and nearly 5× better than gzip. The correlation of 0.96 indicates an almost-linear relationship between circuit depth and the critical reasoning depth in the synthetic channel.

The GSM8K ecological analysis confirms the practical relevance: on 500 real math problems, the correlation between gzip compression length and solution step count is only 0.29, demonstrating that description complexity (what gzip measures) is a weak proxy for computational complexity (what circuit depth captures). This validates the theoretical distinction between Kolmogorov complexity and computational depth.

**Interpretation.** The right question for inference budgeting is not “how long is the problem?” but “how many sequential computation steps does the solution require?” This distinction is practically important because gzip is trivially computable while circuit depth requires estimation—but the 5× MAPE improvement justifies the additional cost.

### 6.4 STAIR Adaptive Allocator

| Method                        | Overall MAPE                       | Low Complexity | High Complexity |
|-------------------------------|------------------------------------|----------------|-----------------|
| Smooth log-concave (baseline) | $72.1\% \pm 0.2$                   | 51.0%          | 92.6%           |
| <b>STAIR (proposed)</b>       | <b><math>47.1\% \pm 4.4</math></b> | 68.4%          | <b>27.6%</b>    |

| Method               | Overall MAPE | Low Complexity | High Complexity |
|----------------------|--------------|----------------|-----------------|
| Relative improvement | <b>34.7%</b> | −34.1%         | <b>70.2%</b>    |

STAIR achieves a 35% relative improvement in overall MAPE, but the gains are entirely concentrated in the high-complexity regime (MAPE: 27.6% vs 92.6%, a 70% improvement). On low-complexity problems, STAIR actually *underperforms* the baseline (68.4% vs 51.0%), because the staircase fitting procedure is less stable when critical depths are near the lower end of the budget grid.

This asymmetry has practical implications: STAIR is most valuable precisely where it is most needed—on hard problems where the baseline’s uniform prediction is maximally wrong. A hybrid allocator that uses the baseline for easy problems and STAIR for hard problems (identified via gzip tercile) would capture the best of both approaches.

## 7. Discussion

### Per-Problem Scaling Is Messier Than Advertised

The smooth log-concave narrative for test-time compute scaling is a useful approximation for system-level budgeting but obscures critical per-problem heterogeneity. Our results show that 41% of per-problem curves are better described as discrete transitions, model identity matters as much as task identity for elbow location, and MI non-monotonicity is the rule rather than the exception. The practical implication is that uniform compute allocation leaves significant efficiency on the table, particularly for high-complexity problems.

### Circuit Depth as the Fundamental Complexity Measure

The dramatic superiority of circuit depth ( $\rho = 0.96$ ) over gzip compression length ( $\rho = 0.38$ ) as an elbow predictor suggests a reframing of the theoretical foundations. The relevant notion of task complexity for reasoning scaling is not algorithmic (Kolmogorov) complexity—which measures description length—but *computational* complexity, which measures the number of sequential steps required for a solution. This aligns with recent work connecting scaling laws to the depth of learned circuits (Li et al., 2024).

### Reasoning Overshoot: A Universal Phenomenon

The 97–98% MI non-monotonicity rate was our most surprising finding. While the magnitude is partly an artifact of the synthetic channel’s planted non-monotonicity, the ubiquity suggests that reasoning overshoot—where extended thinking actively degrades accuracy—is a fundamental property of autoregressive generation, not

an edge case. This has implications for process reward models, which must account for reasoning steps that are not merely uninformative but actively harmful.

## Limitations

1. **Synthetic channel.** Our results are obtained on a synthetic reasoning channel with known parameters. Validation on real LLM inference (e.g., Llama-3 on GSM8K with varying token budgets) is required to confirm that the qualitative findings transfer.
  2. **Two model configurations.** Broader model diversity (architecture, scale, training distribution) is needed to establish the generality of model-dependent elbow divergence.
  3. **Budget grid resolution.** The 8-point exponential budget grid ( $t \in \{4, \dots, 512\}$ ) may miss fine-grained staircase structure, potentially underestimating the BIC win rate.
  4. **Gzip as a bucketing proxy.** While gzip is insufficient for precise elbow prediction, its use for difficulty bucketing in the STAIR allocator is validated only on the synthetic channel.
  5. **Circuit depth estimation.** The practical utility of circuit depth depends on developing cheap estimators—a non-trivial research problem in itself.
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## 8. Conclusion

We introduced STAIR, a framework for analyzing and exploiting the per-problem structure of test-time compute scaling in LLM reasoning. Our key findings are:

- Per-problem scaling curves exhibit meaningful discrete structure (41% staircase by BIC), though the smooth model remains competitive for the majority of problems.
- Model identity and MI non-monotonicity are first-order effects that the channel coding analogy fails to capture.
- Circuit depth is the right complexity measure for inference budgeting ( $\rho = 0.96$ , 10.7% MAPE), dramatically outperforming Kolmogorov complexity proxies.
- The STAIR allocator achieves 35% MAPE reduction on high-complexity problems, where uniform allocation is most wasteful.

These results motivate a shift from population-level smooth bounds to per-problem discrete models for test-time compute allocation, and from description complexity to computational complexity as the fundamental task difficulty parameter.

**Future work.** (1) Validate on real LLM inference at multiple scales (7B–70B). (2) Develop learned circuit depth estimators from problem text. (3) Extend the framework to multi-turn and agentic reasoning,

where compute allocation decisions are sequential. (4) Investigate whether the staircase fraction increases with model scale (as models become more precise reasoners).

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## Appendix A: Experimental Details

**Compute cost.** All experiments ran on a single NVIDIA RTX A4500 GPU (RunPod, \$0.19/hr). Total wall-clock time: 355 seconds. Total cost: < \$0.02.

**Reproducibility.** Code, data generation, and analysis scripts are available at the paper repository. All random seeds are fixed (base seed: 42). Results are deterministic given the same numpy random generator state.

**Statistical tests.** All reported means and standard deviations are computed across 5 independent seeds. Per-condition results are available in the supplementary material.